

CHAPTER 6

CONCLUSIONS AND FUTURE WORK

6.1 Overview and rationale

Over the preceding chapters, this thesis combined experimental tools and mechanistic modeling to dissect how stomata integrate hydraulic and biochemical signals from soil to leaf to regulate water loss under water stress. In maize, a hydropassive model informed by AquaDust measurements[1, 2, 3] and a *slac1* stomatal mutant[4] quantified the contribution of pure hydraulics to stomatal regulation and highlighted further the role of a large outside-xylem (OXZ) resistance that can dominate non-stomatal limitations to transpiration. In Arabidopsis, a HP-HA model of guard-cell-autonomous abscisic acid (ABA) synthesis and signaling showed that ABA autoregulation mediated through the PYR-PP2C-OST1-SLAC1 core signaling module[5, 6] can generate both damped and persistent oscillations in stomatal conductance and the hysteretic “stress memory” observed during drought and recovery. In cotton, AquaDust-based characterization of OXZ conductance and g_s/g_{oxz} across genotypes and vapor pressure deficit (VPD) demonstrated that a C3 crop can exhibit measurable intercellular airspace undersaturation while remaining predominantly stomata-limited. Finally, a Richards-equation soil model coupled to AquaSheet measurements showed that standard soil hydraulic parameterizations can fail dramatically in the extremely dry regime relevant to severe drought.

In what follows, I organize future work into four interrelated themes: species-specific integration of hydropassive and ABA-mediated hydroactive modules; coupling ABA-driven stomatal regulation with the OnGuard platform

and $A-g_s$ modeling; extending the HP–HA framework to Ca^{2+} , ROS, and long-distance signaling; and coupling soil drying with stomatal regulation to form a unified model of the soil–plant–atmosphere continuum (SPAC).

6.2 Species-specific unified hydropassive–hydroactive models

A key limitation of the current work is that the hydropassive model is parameterized and tested primarily in maize, whereas the ABA HP–HA model is built around *Arabidopsis* data and genetic resources. This division was advantageous for leveraging species-specific tools such as AquaDust and maize *slac1* for hydraulics, *Arabidopsis* ABA mutants for understanding hydroactive stomatal regulation. But it leaves open how hydraulics and ABA interact within a single species to shape stomatal responses under water stress. The next step is to construct a unified HP–HA model in a single species, such as maize or cotton, in which both hydraulic and ABA modules are jointly constrained by genotype-resolved data.

In maize, the existing hydropassive model treats mesophyll, subsidiary cells, and guard cells as lumped elements connected by hydraulic resistances and capacitances, with g_s proportional to the turgor difference between guard and subsidiary cells. The model is currently calibrated against *slac1* data, where HA regulation is genetically suppressed, and then used to infer the implicit HA contribution in wildtype. Future work could embed the ABA autoregulation and signaling module developed for *Arabidopsis* directly into this maize HP framework, creating a species-specific HP–HA model in which guard-cell ABA synthesis, ABA positive feedback, and catabolism via CYP707A-family en-

zymes control SLAC1 activation and guard-cell osmotic potential. Such a model would explicitly connect the hydraulic variables measured by AquaDust (xylem and OXZ water potentials) to ABA dynamics and would allow quantitative partitioning of maize drought responses into OXZ hydraulics versus ABA feedback under realistic field-like conditions.

In cotton, the current AquaDust results show that despite measurable OXZ undersaturation, g_s/g_{oxz} remains small, keeping cotton in a stomata-dominated regime across the sampled $\psi_{xyl} - VPD$ space. A unified HP-HA model calibrated on wildtype and GhSLAC1 mutants could test whether species-specific differences in guard-cell elasticity, OXZ vulnerability curves, or ABA feedback can explain why cotton rarely shifts into a strongly non-stomatal regime. Comparative modeling between maize and cotton would directly test the hypothesis that differences in OXZ vulnerability and HA gains underpin species contrasts in intrinsic water-use efficiency.

6.3 Coupling ABA-driven stomatal regulation with OnGuard and $A-g_s$ modeling

At present, the Arabidopsis ABA HP-HA model in this thesis resolves ABA autoregulation, PYR-PP2C-OST1-SLAC1 signaling, and coarse-grained guard-cell electrophysiology, while treating Ca^{2+} implicitly and focusing on K^+ and Cl^- fluxes. The OnGuard platform developed by Blatt and colleagues provides a complementary, guard cell membrane electrophysiology-focused framework. The original OnGuard implementation and its companion systems-dynamics analysis resolve plasma membrane and tonoplast ion channels and pumps,

including Ca^{2+} handling, detailed voltage dynamics, and metabolism of major osmotica under changing light and CO_2 [7, 8]. Subsequent extensions (often referred to as OnGuard2) bridged guard-cell to leaf scales and introduced a “relative-water-feed factor” to represent leaf water status, allowing predictions of transpiration responses to humidity, temperature, and water availability without explicitly resolving the upstream hydraulic circuit[9, 10].

A logical future direction is to integrate these approaches by replacing OnGuard’s phenomenological “hard-wired” ABA input (as in OnGuard3e[11]) with a mechanistic ABA autoregulation module driven by guard-cell turgor and by coupling its electrophysiology to a hydraulically explicit hydropassive (HP) module. In such a hybrid ABA–OnGuard–HP model, step changes in VPD or ψ_{xyl} would first alter guard-cell turgor via the HP module, triggering stress-dependent ABA synthesis $V_{stress}(P_g)$, ABA positive feedback, and catabolic negative feedback via A8H; ABA would then bind PYR receptors, inhibit PP2C, activate OST1, and thereby modulate SLAC1 and related channels within the OnGuard core, leading to changes in membrane potential and guard-cell osmotic potential. Because the HP module explicitly resolves xylem and SSC apoplastic water potentials, this framework would link AquaDust-measured ψ_{xyl} and ψ_{ssc}^{apo} directly to ABA dynamics, Ca^{2+} signaling, and ion transport, rather than treating water status as an external “relative-water-feed” factor. Existing OnGuard models have focused primarily on light and CO_2 responses and, notably, revealed a “carbon memory” whereby guard-cell endomembrane Ca^{2+} -ATPases lead to carbon hysteresis[12]. By contrast, the ABA HP–HA model developed in this thesis predicts a complementary “water-stress memory” arising from ABA autoregulation and its bistable response to xylem water potential during drought and recovery. Integrating these two lines of work would

enable mechanistic exploration of how carbon and water-stress memories interact, for example, how a history of fluctuating CO_2 and light, encoded via Ca^{2+} -dependent feedback, combines with a history of drought, encoded via ABA feedback, to shape subsequent stomatal responses, photosynthesis, and intrinsic water-use efficiency.

Embedding the integrated ABA–OnGuard–HP module into leaf-scale A – g_s models would provide a pathway from guard cell-scale signaling to canopy-scale carbon and water fluxes. At the leaf and canopy scales, most photosynthesis-transpiration models still treat stomatal conductance using empirical formulations such as Ball–Berry[13] or Medlyn[14] models, which relate g_s to photosynthetic assimilation, atmospheric CO_2 , and VPD via fitted parameters and assume that stomata present the dominant resistance to water vapor flux, thereby neglecting losses of outside-xylem conductance and intercellular airspace undersaturation. The maize and cotton chapters in this thesis demonstrate that such assumptions can fail under moderate drought. In maize, g_s/g_{oxz} can approach 0.3–0.5, causing substantial intercellular airspace undersaturation and non-stomatal control of transpiration, whereas in cotton, g_s/g_{oxz} remains small but ψ_{ssc}^{apo} can still fall several MPa below ψ_{xyl} . Future work should therefore replace empirical g_s models in A – g_s frameworks with mechanistic predictions from ABA–OnGuard–HP models that account for both hydraulic and biochemical state variables and use these models to explore how different combinations of OXZ vulnerability, ABA feedback gains, and guard-cell electrophysiological properties shape the co-evolution of A , E , and A/E during soil drying and rewetting.

By extending OnGuard’s original scope beyond light and CO_2 to encom-

pass ABA-mediated water stress responses, and by anchoring the model in explicit leaf and soil hydraulics, this future work would move toward truly ecophysiology-ready stomatal models. Such models could predict species-specific responses under a wider range of environmental conditions including fluctuating VPD, CO_2 , light, and soil water status and provide mechanistic insight into water-use efficiency from both the water and assimilation perspectives.

6.4 Extending the HP–HA framework to Ca^{2+} , ROS, and long-distance signaling

The current ABA HP–HA model treats Ca^{2+} implicitly, black-boxing its effects into coarse-grained rate laws for ion fluxes and membrane voltage while holding symplastic pH fixed. Ca^{2+} dynamics in guard cells, however, form an important secondary messenger system that amplifies and encodes information about ABA[15]. These Ca^{2+} signals act through a kinase branch parallel to the OST1 branch that converges on SLAC1[16, 17, 18, 19]. In the Ca^{2+} -dependent branch, CPKs directly phosphorylate the SLAC1 N-terminus and are themselves negatively regulated by PP2C phosphatases such as ABI1. The OSCA family of hyperosmolality-gated, Ca^{2+} -permeable channels provides a mechanosensing entry point into this network. OSCA1 mediates osmotic-stress-evoked cytosolic Ca^{2+} increases needed for osmosensing, and *osca1* mutants show attenuated regulation of transpiration under osmotic stress[20]. Because our current $V_{stress}(\delta P_g)$ formulation is agnostic about which molecular sensors convert turgor into a biochemical signal, incorporating an OSCA-like Ca^{2+} module where guard-cell

turgor changes computed by the HP module drive mechanosensitive Ca^{2+} influx into the CPK branch would allow Ca^{2+} -dependent effects on SLAC1 and related targets to be captured explicitly and analyzed via bifurcation tools (Appendix B), yielding testable hypotheses in *osca1* and *cpk* mutants.

A critical dimension of such an extension is the integration of guard-cell Ca^{2+} signaling with long-distance hydraulic signals and with the experimental tools needed to constrain the resulting models. Recent work by Bacheva et al.[21] introduces a unifying poroelastic framework that captures both local and systemic hydraulic responses to wounding and predicts $\psi_{xyl}(t)$ relaxation dynamics that depend on transpiration. Feeding these poroelastic $\psi_{xyl}(t)$ trajectories into the HP-HA- Ca^{2+} model would create a mechanistic bridge between whole-plant hydraulic states and guard-cell responses, in which Ca^{2+} signals and ABA synthesis respond explicitly to plant-scale hydraulics, enabling predictions for soil drought (slow, abiotic), wounding (fast, biotic), or their combinations. Testing these mechanistic models requires tools that can simultaneously report Ca^{2+} dynamics and physiological output with sufficient spatial and temporal resolution. Genetically encoded Ca^{2+} reporter plants now make it possible to visualize stimulus-specific Ca^{2+} transients *in vivo* across a range of timescales and spatial scales[22, 23, 24]. Combining these Ca^{2+} reporters with AquaDust measurements and LI-COR-based gas exchange would create a phenotyping platform in which molecular Ca^{2+} signals are directly correlated with the hydraulic and physiological state variables that drive the model, providing the observational constraints needed for parameter inference and validation.

Reactive oxygen species (ROS) are another major axis of guard-cell signaling that the current HP-HA model does not resolve explicitly. ABA activates the

NADPH oxidases AtRBOHD and AtRBOHF in guard cells, generating apoplastic H_2O_2 bursts that in turn activate Ca^{2+} -permeable channels, modulate SLAC1, and drive stomatal closure[25, 26]. More recently, work from the Woodson laboratory at the University of Arizona has shown that chloroplast-derived ROS signals form a part of specialized signaling networks[27, 28]. These findings point to a richer ROS landscape in guard cells than is currently captured and motivate the inclusion of a minimal ROS module in the HP-HA framework, enabling exploration of combined stress scenarios such as heat plus drought, where ROS production is elevated independently of ABA, and prediction of how such stresses modify stomatal responses. Genetically encoded ROS reporters of cellular ROS[29, 30] could be combined with the Ca^{2+} imaging and AquaDust platforms described above to simultaneously monitor ROS, Ca^{2+} , and hydraulic state. This multi-signal approach would provide rich observational constraints for the extended model and help dissect the specificity with which each molecular mechanism contributes to stomatal regulation, a central question in the rapidly expanding field of long-distance signaling in plants[31, 32].

6.5 Coupling soil drying, root uptake, and stomatal regulation

The soil chapter shows that a standard 1D Richards-equation model with van Genuchten retention and Philip de Vries vapor conductivity can be tuned to fit moderate soil-drying data but fails once soil dries beyond -20 to -30 MPa at the surface. This breakdown reflects extrapolation of constitutive soil relations beyond their calibration range and omission of heterogeneity effects and underscores that soil hydraulic behavior in the extremely dry regime remains poorly

constrained. Independent work using neutron radiography and hydraulic modeling indicates that stomatal downregulation may be initiated by hydraulic limitations at the root–soil interface rather than by xylem embolism or leaf-level events[33, 34].

A future direction is to couple the Richards-based soil model to a root water uptake module and then to the HP–HA stomatal model, creating an integrated soil-root-xylem-leaf framework constrained by AquaSheet and AquaDust measurements. AquaSheet-derived $\psi_{soil}(z, t)$ profiles would feed into a root uptake model that accounts for root length density and rhizosphere conductivity[35, 33], which in turn provides xylem boundary conditions to the leaf HP module. The HP–HA model would then predict g_s , ψ_{ssc}^{apo} , ABA, Ca^{2+} , and ROS dynamics, closing the loop with transpiration and feeding back on soil drying. This integrated framework would allow studying the SPAC under different soils and atmospheric forcing.

6.6 Concluding perspective

The modeling work in this thesis highlights the importance of rigorous parameter inference and identifiability analysis. The maize hydropassive model uses MCMC with Bayesian learning to quantify posterior distributions and parameter correlations, revealing which parameters are tightly constrained and which remain weakly identified by the steady-state data. The ABA HP–HA model similarly relies on non-dimensionalization and sequential fitting to isolate key dimensionless groups governing oscillations and hysteresis. Future work should generalize these inference pipelines to include additional observables such as

Ca²⁺ and ROS data.

In this thesis, we develop a mechanistic framework that reconciles five decades of stomatal physiology and genetic evidence into a unified, quantitative model of stomatal regulation. The fact that such a synthesis was missing for so long underscores the need for shared tools so similar gaps do not persist in the future. One avenue is to develop open-source inference tools that allow other laboratories to fit HP-HA models to their own gas-exchange, hydraulic, and signaling datasets, thereby lowering the barrier to mechanistic modeling of stomata. To that end, we demonstrate how hosting interactive dashboards for model exploration and parameter fitting, for example, the publicly available stomatal conductance modeling dashboards at <https://desai-sahil.github.io/stomatal-conductance-model/>, can serve as powerful tools for sharing models, visualizing fits, and engaging users with varying levels of quantitative expertise. In the current landscape of advanced AI-assisted coding and analysis tools, such dashboards provide a flexible way to couple transparent mechanistic models to user-friendly interfaces, enabling rapid iteration on hypotheses and a more accessible pathway from raw experimental data to rigorously inferred parameters.

The future directions outlined above aim to extend this framework toward species-specific, unified models that integrate hydraulics, ABA, Ca²⁺, ROS, and photosynthesis across scales from soil to canopy. Pursuing these projects will require coordinated advances in experiments, modeling, and integration with genetics, breeding, and field-scale measurements, but offers a path toward predictive platforms that can inform the design of crops and management strategies resilient to a warming and increasingly water-limited world.

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